Lab 07:

	Outlook	Temp	Humidity	Windy	Rain
	OL	T	H	W	R
1	Sunny	Hot	High	Weak	No
2	Sunny	Mild	High	Strong	No
3	Cloudy	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Cloudy	Mild	Normal	Strong	Yes
8	Sunny	Cool	Normal	Weak	No

Entropy of Prime Att.

- Rain:

Y N

$$-\left(\frac{4}{8}\log_2\left(\frac{4}{8}\right) + \frac{4}{8}\log_2\left(\frac{4}{8}\right)\right)$$

$$= \boxed{1}$$

Signature

No.

Entropy of Non-Prime Attributes:

- Outlook:

- Sunny:

Y N

$$\left(0 + \frac{3}{3}\log_2\left(\frac{1}{1}\right)\right)$$

$$= \boxed{0}$$

- Cloudy:

Y N

$$\left(\frac{2}{2}\log_2\left(\frac{2}{2}\right) + 0\right)$$

$$= \boxed{0}$$

Rainy:

Y N

$$-\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)$$

$$= \boxed{0.918}$$

Signature

No.

Weighted Entropy:-

- Sunny = 0 , Cloudy = 0

- Rainy = $0.918 \times \frac{3}{8}$

Total = 0.344

IG of Outlook

= Entropy of Rain - Total weighted Entropy

= 0.656 //

* Temperature

- Hot :-

$$= -\left(\frac{1}{2}\log_2\left(\frac{1}{2}\right) + \frac{1}{2}\log_2\left(\frac{1}{2}\right)\right)$$

= 1

- Mild :-

$$= -\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)$$

= 0.918 //

- Cool

$$= -\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right)$$

= 0.918

Weighted Entropy:-

- Hot :-

$$1 \times \frac{2}{8} = 0.25$$

- Mild :-

$$= 0.918 \times \frac{3}{8} = 0.344$$ //

- Cool :- = 0.344 //

Total = 0.938

IG of Temperature:

= 1 - 0.938

= 0.062

- Humidity:

- High

$$-\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right)\right)_{Y} + \left(\frac{2}{4}\log_2\left(\frac{2}{4}\right)\right)_{N}$$

= 1

- Normal:

$$-\left(\frac{2}{4}\log_2\left(\frac{2}{4}\right)\right)_{Y} + \left(\frac{2}{4}\log_2\left(\frac{2}{4}\right)\right)_{N}$$

= 1

Weighted Entropy:

- High:

$$\frac{1 \times 4}{8} = 0.5$$

Normal:

$$\frac{4}{8} \times 1 = 0.5$$

Total = 1

IG of Humidity:

1 - 1 = 0

Windy:

Weaker

$$\frac{3}{5}\log_2\left(\frac{3}{5}\right)_{Y} + \frac{2}{5}\log_2\left(\frac{2}{5}\right)_{N}$$

= 0.971

Strong

$$\frac{1}{3}\log_2\left(\frac{1}{3}\right)_{Y} + \frac{2}{3}\log_2\left(\frac{2}{3}\right)_{N}$$

= 0.918

Date

Weighted Entropy:

-Weak:

$0.971 \times \frac{5}{8} = 0.607$

-Strong

$\frac{3}{8} \times 0.918 = \boxed{0.344}$

Total = $\boxed{0.951}$

IG of Wind:

$1 - 0.951 = \boxed{0.049}$

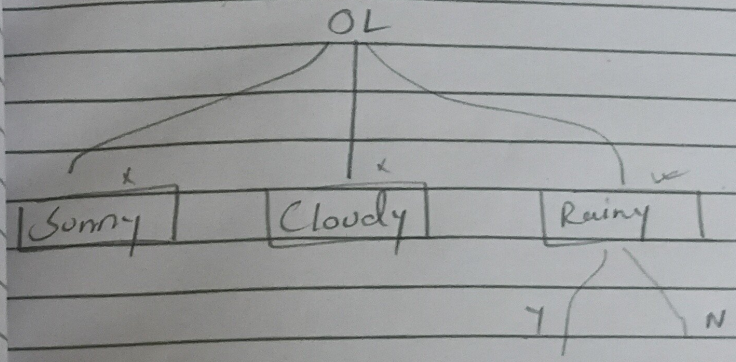
Aff	Entropy	IG
OL	0.344	0.656
T	0.938	0.062
H	1	0
W	0.951	0.049

Signature

No.

Date

OL has the highest IG



Data Set:

OL	T	H	W	R
Rainy	Mild	Normal	Weak	Y
Rainy	Cool	Normal	Strong	Y
Rainy	Cool	Normal	Strong	N

Signature

No.